

EduGenie – Google Gemini Powered Learning Assistant

Project Design Phase Document

Team Leader: P. Abirami

Team Members: P. Abinaya M.S. Harini V. Nithishree

Email: abigeetha2007@gmail.com

1. Introduction

The design phase converts the EduGenie requirements into a structured system design. It describes the overall architecture, modules, user interface, data flow, interaction between components, and design decisions required to develop the application.

2. Design Objectives

- Create a simple and student-friendly interface.
- Provide a clear flow from user question to AI-generated response.
- Separate the user interface, application logic, and AI service interaction.
- Make the design easy to maintain and extend.
- Support future educational features such as quizzes and personalized learning.

3. Proposed System Architecture

EduGenie can be designed as a layered application. The user interacts with the presentation layer. The application layer receives the request, prepares the prompt, communicates with Google Gemini, and returns the generated response to the interface.

Layer	Main Responsibility
Presentation Layer	Displays the EduGenie interface, input area, buttons, and AI response.
Application Layer	Processes user input, manages requests, prepares prompts, and handles responses.
AI Integration Layer	Connects the application to Google Gemini and receives generated content.
Data / Configuration Layer	Stores configuration information and, if required, learning or session data.

4. High-Level System Flow

User → EduGenie Interface → Request Processing → Gemini API → AI Response → Response Processing → User Interface

5. Module Design

Module	Purpose
User Interface Module	Provides the screen through which students enter questions and view responses.
Input Processing Module	Validates and prepares the user's question or topic.

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Prompt Management Module	Creates a suitable prompt for the learning task.
Gemini Integration Module	Sends the request to Google Gemini and receives the response.
Response Display Module	Formats and displays the generated answer clearly.
Error Handling Module	Handles empty input, connection issues, service errors, and invalid requests.

6. User Interface Design

The interface should be simple, clean, and focused on learning. The main screen can contain the EduGenie title, a question/topic input area, an action button, and a response section.

UI Component	Design Purpose
Header / Title	Clearly identifies the application as EduGenie.
Question Input	Allows students to type a question or learning topic.
Action Button	Starts the AI request.
Response Area	Displays the generated explanation or study material.
Loading / Status Message	Indicates that the request is being processed.
Error Message	Provides understandable feedback when a request fails.

7. Data Flow Design

Step	Process
1	Student enters a question or topic.
2	The interface sends the input to the application logic.
3	The application validates the input and prepares a learning prompt.
4	The prompt is sent to Google Gemini.
5	Gemini generates a response.
6	The application receives and processes the response.
7	The response is displayed to the student.

8. Input and Output Design

Type	Examples
Input	Question, topic, request for explanation, request for notes, practice-question request.
Processing	Input validation, prompt creation, AI request, response handling.
Output	Explanation, notes, examples, practice questions, or other learning-focused content.

9. Error Handling Design

- Empty question: ask the user to enter a question or topic.
- Network problem: display a message asking the user to check the connection.
- AI service error: display a suitable retry message.
- Invalid request: provide clear guidance to the user.
- Unexpected error: prevent the application from failing silently and show user-friendly feedback.

10. Security Design

- API credentials should be kept private and should not be displayed in the frontend.
- Configuration and secret values should be stored securely.

- User input should be validated before processing.
- The application should avoid exposing internal error details to normal users.

11. Technology Design

Technology / Component	Design Role
Frontend	Provides the interactive EduGenie user interface.
Application Logic	Controls input processing, prompts, API calls, and response handling.
Google Gemini	Provides generative AI capabilities.
Development Environment	Used to write, run, debug, and test the project.
Web Browser / Runtime	Used to access and test the application, depending on implementation.

12. Design for Learning Features

- **Explain:** Provide step-by-step explanations for difficult topics.
- **Summarize:** Convert a topic into concise revision notes.
- **Practice:** Generate questions for self-assessment.
- **Follow-up:** Allow students to ask additional questions about the same topic.

13. Design Principles

- Simplicity – keep the interface easy to understand.
- Consistency – use consistent controls, messages, and layout.
- Accessibility – make text and interactions easy to read and use.
- Maintainability – keep modules organized and separated.
- Extensibility – allow additional AI learning features to be added later.

14. Testing Considerations in Design

- Test normal questions and different types of learning requests.
- Test empty and invalid inputs.
- Test API or network failure scenarios.
- Check whether responses are displayed correctly.
- Test the interface on supported screen sizes or environments.

15. Future Design Extensions

- Voice-based interaction.
- Tamil and multilingual learning interface.
- Student profiles and progress tracking.
- Personalized study-plan module.
- Image-based question and learning support.
- Advanced quizzes and examination preparation.

16. Conclusion

The project design phase provides the technical and structural blueprint for EduGenie. The architecture, modules, interface, data flow, security considerations, and future extensions described here can guide the implementation and testing of the Google Gemini powered learning assistant.